## 6. Project Documentation Phase

### Introduction

The **Project Documentation Phase** is an important part of the **Smart Library Request Workflow in ServiceNow** project. Documentation helps explain how the system is designed, developed, and implemented so that other developers, administrators, or users can understand and reproduce the system if needed.

Proper documentation ensures that the project is easy to maintain, update, and deploy in different environments. In this phase, the project includes two main documentation components: the **Technical Blueprint** and the **Setup Manual**.

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### Technical Blueprint

The **Technical Blueprint** provides a detailed description of the internal structure of the system. It explains how the application works, including the tables used, variables involved, workflows implemented, and how different components interact with each other.

This blueprint helps developers and system administrators understand the logic and design of the Smart Library Request Workflow.

### Variables and Data Fields

The system includes several variables and fields that are used to manage borrowing requests and track book transactions.

Some important variables include:

- **Borrow Date** – The date when a student borrows a book.
- **Return Date** – The expected date when the book should be returned to the library.
- **Book Status** – Indicates whether the book is Available, Issued, or Lost.
- **Request Status** – Shows the status of a borrow request (Pending, Approved, Rejected).

### Example

When a student submits a borrow request, the system records the **Borrow Date** and sets the **Request Status** to *Pending*. Once the librarian approves the request, the **Book Status** changes to *Issued* and the **Return Date** is recorded.

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### Fulfillment Task Use Cases

The system also includes **fulfillment tasks** that help librarians manage requests efficiently. A fulfillment task is automatically generated when a borrow request requires librarian action.

These tasks are assigned to the **Librarian Group**, allowing library staff to review and process requests in an organized manner.

**Example Scenario**

1. A student submits a request for the book *"Database Management Systems."*
2. The system creates a **Fulfillment Task** assigned to the librarian group.
3. A librarian reviews the request and approves it.
4. The system updates the request status and book status accordingly.

This process ensures that requests are handled systematically and that librarians can easily track pending tasks.

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**Setup Manual**

The **Setup Manual** provides a clear and step-by-step guide for recreating the Smart Library Request Workflow in a **ServiceNow Personal Developer Instance (PDI)**. This guide is especially useful for developers or students who want to implement the same solution in their own environment.

The manual explains the order of operations and includes instructions for configuring tables, roles, workflows, and automation features.

**Step-by-Step Setup Process**

**Step 1: Create Roles**

- Create two roles in ServiceNow:
    - Student
    - Librarian

These roles will define user permissions and access levels.

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**Step 2: Create Tables**
Create two custom tables in the ServiceNow database:

1. **u_book** – Stores book details such as title, author, ISBN, and status.
2. **u_borrow_request** – Stores borrow request details including student name, book reference, and request status.

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**Step 3: Add Fields**
Add necessary fields to the tables.

Example fields for **u_book table**:

- Title
- Author
- ISBN
- Category
- Status

Example fields for **u_borrow_request table**:

- Student Name
- Book Reference
- Borrow Date
- Return Date
- Request Status

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**Step 4: Configure Forms**
Design forms for both tables so that users can easily view and enter information.

For example:

- Students use a form to submit borrow requests.
- Librarians use a form to update book details and request statuses.

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**Step 5: Create Workflow Using Flow Designer**
Develop a flow that triggers when a borrow request is approved.

The workflow should:

- Update the book status to **Issued**
- Send a notification to the student

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**Step 6: Add Business Rules and Notifications**
Create business rules that automatically send email notifications when a request is approved or rejected.

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**Step 7: Apply Reference Qualifiers**
Configure reference qualifiers so that students can only see books with the status **Available** when creating a request.

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**Step 8: Test the System**
Perform testing to ensure that:

- Borrow requests are submitted correctly
- Librarian approvals work properly
- Book status updates automatically
- Notifications are sent to students

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**Use of Screenshots**

The setup manual also includes screenshots of key configuration steps, such as:

- Table creation in ServiceNow
- Form design
- Flow Designer configuration
- Notification setup

Screenshots help users follow the instructions more easily and understand the system setup visually.

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**Summary**

The **Project Documentation Phase** ensures that the Smart Library Request Workflow system is well-documented and easy to understand. The **Technical Blueprint** explains the internal structure and workflow of the system, while the **Setup Manual** provides a clear guide for implementing the solution in a ServiceNow environment.

This documentation helps developers, administrators, and future users understand how the system works and enables them to recreate or maintain the application effectively.